

RISK ASSESSMENT MEMO

Organization: BrightLayer Stores

1. FIVE IMPORTANT ASSETS

Asset 1: Employee email accounts

Why it matters: Emails are used by the employees to complete their daily tasks , to communicate between their co workers, and to access company related resources.

Asset 2: Shared computers

Why it matters: Computers are used to store files , browse resources, and to do their daily business activities

Asset 3: Company website

Why it matters: The website represents BrightLayer Stores online and may provide information or services to customers. Website downtime or compromise could affect business operations and reputation.

Asset 4: Cloud storage with invoices

Why it matters: Invoices contain important financial and business information. Losing, changing, or exposing these records could cause financial and operational problems.

Asset 5: User passwords

Why it matters: Passwords protect employee accounts, computers, cloud storage, and other systems. Compromised passwords could provide attackers with unauthorized access.

2. THREE POTENTIAL THREATS

Threat 1: Malicious insider

what is the threat: An employee could intentionally misuse their access to view, modify, or steal company information.

Threat 2: Unauthorized users on public Wi-Fi

what is the threat: An attacker connected to the public Wi-Fi environment may attempt to intercept traffic or gain unauthorized access to company devices and resources

Threat 3: Credential theft**what is the threat:** An attacker could obtain employee usernames and passwords through phishing, credential leaks, or other methods and use them to access company systems

3. THREE VULNERABILITIES

Vulnerability 1: Weak or reused user passwords**Exact weakness:** If employees use simple or reused passwords, attackers may be able to guess or obtain credentials and reuse them against other company systems.

Vulnerability 2: Lack of multi-factor authentication (MFA)

Exact weakness: If email and cloud accounts rely only on passwords, a stolen password may be sufficient for an attacker to access them.

Vulnerability 3: Insecure public Wi-Fi configuration

Exact weakness: Using a public Wi-Fi router without strong wireless security, proper segmentation, or a separate employee network could expose company devices to unauthorized network access.

4. ASSOCIATED RISKS

Risk 1: Cybercriminal + weak/reused passwords:

Asset: employee passwords

Threat: Cybercriminals

Vulnerability: weak/reused passwords

Risk: An attacker who obtains an employee password could access email or cloud services. This could expose confidential information and potentially lead to unauthorized changes or financial loss.

Risk 2 : Phishing attacker + lack of MFA

Asset: employee email accounts

Threat: Phishing attacker

Vulnerability: lack of MFA

Risk: A successful phishing attack could result in stolen employee credentials. Without MFA, the attacker may be able to take over the employee's email or cloud account and access business information.

Risk 3: Unauthorized Wi-Fi user + insecure Wi-Fi configuration

Asset: public Wi-Fi router.

Threat: Unauthorized Wi-Fi user **Vulnerability:** insecure Wi-Fi configuration

Risk: An attacker could gain access to the same network as company devices. This could increase the risk of unauthorized access, traffic interception, malware infection, or disruption of business operations.]

5. SECURITY RECOMMENDATIONS

1. Implement strong, unique passwords for all employee accounts to reduce the risk associated with weak or reused credentials.
2. Enable MFA for email, cloud storage, and other important accounts. This reduces the impact of stolen passwords.
3. Provide phishing-awareness training so employees can recognize suspicious emails, links, attachments, and login requests.
4. Secure and segment the Wi-Fi network using WPA2/WPA3, a strong router administrator password, and a separate network for business devices where possible. This reduces the risk from unauthorized Wi-Fi users.
5. Apply least-privilege access so employees only have access to the files and systems required for their jobs.
6. Keep shared computers and website software patched and updated to reduce the chance that attackers can exploit known vulnerabilities.
7. Maintain regular backups of invoices and important business data so the company can recover if information is deleted, corrupted, encrypted, or otherwise lost.

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